■ `ArcTan`

`ArcTan[`z`]` gives the arc tangent $\tan^{-1}(z)$ of the complex number z.

`ArcTan[`x`, `y`]` gives the arc tangent of $\frac{y}{x}$, taking into account which quadrant the point (x, y) is in.

See notes for `ArcCos`. ■ If x or y is complex, then `ArcTan[`x`, `y`]` gives $-i \log\left(\frac{x+i\,y}{\sqrt{x^2+y^2}}\right)$. When $x^2 + y^2 = 1$, `ArcTan[`x`, `y`]` gives the number ϕ such that $x = \cos\phi$ and $y = \sin\phi$. ■ See page 353. ■ See also: `Arg`.

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